How ChatGPT remembers, and whether long conversations make it worse

A structured narrative evidence review · Research ID chatgpt-memory-long-conversations-2026-11 · Searches and documentation consulted 3 November 2026 · Product descriptions reflect OpenAI's pages on that date and may differ by plan and change later.

The short answer

ChatGPT does not have one memory but several, and they work very differently. Inside a single conversation the model re-reads the conversation text each time it answers, but only as much as fits in a limited *context window*. Across conversations it relies on separate features: **saved memories** (a short "notepad" that OpenAI says is always considered), **reference chat history** (a selective lookup that OpenAI says does not keep every detail), and, inside Projects, **project memory** plus project files and instructions. Which of these are switched on, and which boundaries a Project sets, largely decides what can be recalled. This explains, without any malfunction, how a detail from weeks ago can be remembered (it was turned into a saved memory) while something important from yesterday is missed (it existed only in a past chat that the selective lookup did not bring back).

On the second question, the research is fairly consistent: across many models, accuracy tends to fall as inputs get longer, information in the middle of long inputs is used less reliably, and models perform clearly worse when a task is built up over several turns than when it is stated all at once. But the effect depends strongly on the kind of task, it is measured mostly in laboratory tests rather than in ChatGPT itself, and part of a perceived decline can come from other causes: harder later requests, ordinary randomness, and model updates.

1. The kinds of "memory" and why recall seems inconsistent (Q1)

The intuitive picture. A language model does not store your conversation inside itself. Every time it answers, the app sends it a bundle of text to read: hidden instructions, any memory notes, relevant material from files or older chats, and the current conversation. Whatever is in that bundle can influence the answer; whatever is not, the model simply cannot see. "Memory" features are different ways of deciding what goes into the bundle. The bundle's size is limited and measured in *tokens* (word-pieces; roughly three-quarters of an English word each).

Mechanism	What it holds	When it is used (per OpenAI)	Main limit
Current conversation (context window)	The text of this chat	Every reply, as long as it fits	Window size depends on plan and model; very long chats exceed it
Saved memories	Short facts you asked it, or it chose, to remember	Always considered in future replies unless deleted; stored separately from chat history	Brief notes, not full records; can become stale or contradictory
Reference chat history / memory summary	A continually updated synthesis of past chats	When relevant, if the setting is on	Does not remember every detail; the summary may not list everything
Project files and instructions	Documents and rules you add to a Project	In chats inside that Project	File limits per plan (5, 25 or 40 files)
Project memory	Context from other chats in the same Project	Prioritised in Project chats; needs memory settings on	No viewable list; boundaries depend on mode and plan

Documented facts. OpenAI's Memory FAQ (consulted 3 Nov 2026) describes saved memories as details you directly told ChatGPT to remember, treated like custom instructions and always considered unless deleted; they are kept apart from chat history, so deleting a chat does not delete a memory made from it. It describes the newer system as a continually updated synthesis of past chats, shown as a "memory summary" that may not include everything ChatGPT remembers, and it advises using saved memories for anything that must always be kept, because past-chat reference does not keep every detail. Temporary Chats neither use nor create memories. Memory controls sit under Settings › Personalization › Memory (the exact menu path varies by plan and app version), and a "sources" icon under a reply can show which memories, chats or files personalised it, though OpenAI says it may not

show every factor.

For Projects, OpenAI's help page says each Project has either **default** or **project-only** memory. Project-only memory ignores your saved memories and other chats outside the Project, but can draw on other chats in the same Project. With default memory, outside the Enterprise and Edu plans, Project chats can also draw on non-Project conversations. Sharing a Project switches it permanently to project-only. Crucially, project memory only works if **Reference saved memories** and **Reference chat history** are both enabled (on most plans), and there is no list of project memories to inspect; the only way to make ChatGPT ignore a chat is to delete it or move it out. Third-party guides from August 2026 report that eligible unshared Projects can now switch memory mode after creation, which conflicts with older wording on the help page; this is unresolved here.

Why weeks-ago recall can beat yesterday (inference from the documentation, not tested). Several documented paths produce exactly the reported pattern. (1) An old detail that became a saved memory or entered the memory summary is injected into every chat; a recent detail that lives only in yesterday's conversation depends on a selective lookup that may not retrieve it. (2) If “Reference chat history” is off, other chats are invisible, even inside a Project, while saved memories still work. (3) A project-only Project cannot see chats outside it, and a chat in another Project or a Temporary Chat cannot be reached. (4) Within one very long conversation, early material may no longer fit in the window. OpenAI does not publish how the lookup ranks what is “relevant”, or exactly what happens when a conversation exceeds the window, so the precise cause in a given case cannot be determined from the outside.

Context window size. OpenAI lists different windows by plan and by model type (fast “instant” models versus “reasoning” models). Third-party summaries dated 2026 report figures from roughly 16–54 thousand tokens for instant models on lower plans to 256–400 thousand for reasoning models on higher plans, but they disagree with one another and the current OpenAI table could not be retrieved directly, so no exact figure is given here.

2. Do answers really get worse in long conversations? (Q2)

No published study found here tests the ChatGPT app itself over long real conversations. The evidence comes from controlled experiments on models, including several OpenAI models that power or powered ChatGPT. Two separate effects are relevant: *long input* (lots of text in the window) and *many turns* (a task assembled piece by piece).

Effect A: long inputs

- **Lost in the middle** (Liu et al., peer-reviewed, TACL 2024). On question answering over many documents, models used information best when it sat at the start or end of the input and worst when it was in the middle, a U-shaped pattern; accuracy also fell as inputs grew. Tested models date from 2023, including GPT-3.5.
- **NoLiMa** (Modarressi et al., ICML 2025). When the question does not share wording with the hidden fact, so the model must make a connection rather than spot a matching phrase, 11 of 13 models fell below half their short-input score at 32,000 tokens; GPT-4o dropped from 99.3% to 69.7%. The authors note newer models such as GPT-4.1 do better, but still decline well before their advertised limits.
- **Length alone** (Du et al., EMNLP 2025 Findings). Even when five models could reproduce the needed information perfectly, accuracy on maths, question-answering and coding fell by 13.9% to 85% as filler was added, and drops persisted when the filler was blank space.
- **Context Rot** (Hong et al., Chroma technical report 2025, not peer-reviewed). Eighteen recent models became less reliable as input grew, even on simple tasks such as repeating text; distracting, similar-looking content made it worse.

Effect B: many turns

The most directly relevant study is **LLMs Get Lost in Multi-Turn Conversation** (Laban et al., arXiv preprint 2025; a third-party note lists it as an ICLR 2026 paper, not verified here). Over 200,000 simulated conversations with 15 models, including GPT-4o, GPT-4.1 and o3, the same tasks revealed one piece per turn scored on average 39% lower than when stated in one message. The drop appeared

already with two turns. Best-case ability fell only modestly; what grew sharply (+112%) was *unreliability*: the gap between good and bad runs of the same conversation. Models guessed missing details early, attempted final answers too soon, and then leaned on their own earlier wrong attempts. When all pieces were gathered into one message, performance returned to about 95% of the original.

Boundary conditions and counterevidence

- **Not every task degrades.** In the same study, a translation task, where each turn adds a separate chunk, showed no drop. Degradation was largest when each new piece changes the whole answer (code, calculations, queries). Literal "find the sentence" tests are near-perfect for modern models; harder reasoning is where length hurts.
- **Length versus difficulty.** The Chroma authors note that longer tasks are often also harder, which makes the pure effect of length difficult to isolate; Du et al. address this but used only five models.
- **Simulated, not real users.** Laban et al. used an AI to play the user and say real conversations are likely messier, so they consider their numbers underestimates; this cannot be checked from the outside.
- **Models change.** Chen, Zaharia and Zou (Harvard Data Science Review 2024) found that GPT-4's accuracy on some tasks fell sharply between March and June 2023 while improving on others. A decline noticed over weeks may reflect an update rather than conversation length.

Synthesis. The impression that very long chats get worse is supported as a real tendency in models of the kind ChatGPT uses, through two mechanisms: crowded, long context and the accumulation of early wrong turns. Neither study quantifies how often this happens in everyday ChatGPT use, and plausible alternative causes (harder late requests, chance variation that is high in multi-turn settings, and model updates) can produce the same impression. A single disappointing answer is weak evidence; a repeated pattern on the same kind of task, late in long chats, is more consistent with the research.

3. Habits the evidence supports (Q3)

These are conditional options, each tied to its source. None has been tested on ChatGPT end to end.

If you want…	Option	Basis
A fact kept reliably across chats	Ask ChatGPT explicitly to remember it (saved memory), or put it in custom or Project instructions or a Project file	OpenAI: saved memories are always considered; past-chat reference does not keep every detail
To know what is switched on	Check Settings › Personalization › Memory for both "Reference saved memories" and "Reference chat history"; check the Project's memory mode; use the sources icon	OpenAI Memory FAQ and Projects help page
Recall across chats in one Project	Keep related chats in the same Project; for isolation choose project-only; delete or move chats you want ignored	OpenAI Projects help page
Better answers when a chat drifts	Ask for a summary of all requirements so far, then start a new chat with that summary as one complete message	Laban et al.: one consolidated message restored ~95%; repeating information each turn helped partly
Fewer errors with long material	Give only the material that is needed; put key instructions at the start or end; ask the model to quote the relevant passage first	Liu et al.; Du et al. (reciting evidence first helped modestly)
To judge whether quality really fell	Retry the same request in a fresh chat before concluding	Laban et al.: large run-to-run variation; Chen et al.: model drift

4. Limitations of this review

This is a structured narrative review, not a systematic one. OpenAI's help pages describe intended behaviour, not measured performance; two key pages blocked automated retrieval and were read through search-result extracts and an official mirror, so wording may be incomplete. Current context-window figures could not be confirmed from OpenAI and conflicting third-party numbers were omitted. Research findings come mainly from API or open models, often older than today's ChatGPT models, and from short, structured tasks; applicability to everyday ChatGPT use is inferred.

Because of the strict five-page limit, the review omits other long-context benchmarks, studies of mitigation by summarising or compressing chat history, and plan-by-plan differences in Enterprise and Edu settings. The requester's own settings and conversations were not examined, as agreed.

References and documentation consulted (accessed 3 November 2026)

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Chen, L., Zaharia, M. & Zou, J. (2024). How Is ChatGPT's Behavior Changing Over Time? *Harvard Data Science Review*. doi:10.1162/99608f92.5317da47; arXiv:2307.09009.

Third-party guides used only for flagged, unverified points (context-window figures; Project memory-mode switching): ai-toolbox.co (Sept 2026), chatai.guide (May 2026), memx.app (Aug 2026), elephas.app (Aug 2026).